

CAT 2025

MBA FASTRACK BATCH



Lecture - 04

Data Interpretation

Advanced Data Sets 02

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TOPICS

to be covered

- 1 DI Sets based on Scattered Charts
- 2 DI Sets based on Bubble Charts ✓✓
- 3 DI Sets based on Candle Stick Charts ✓✓



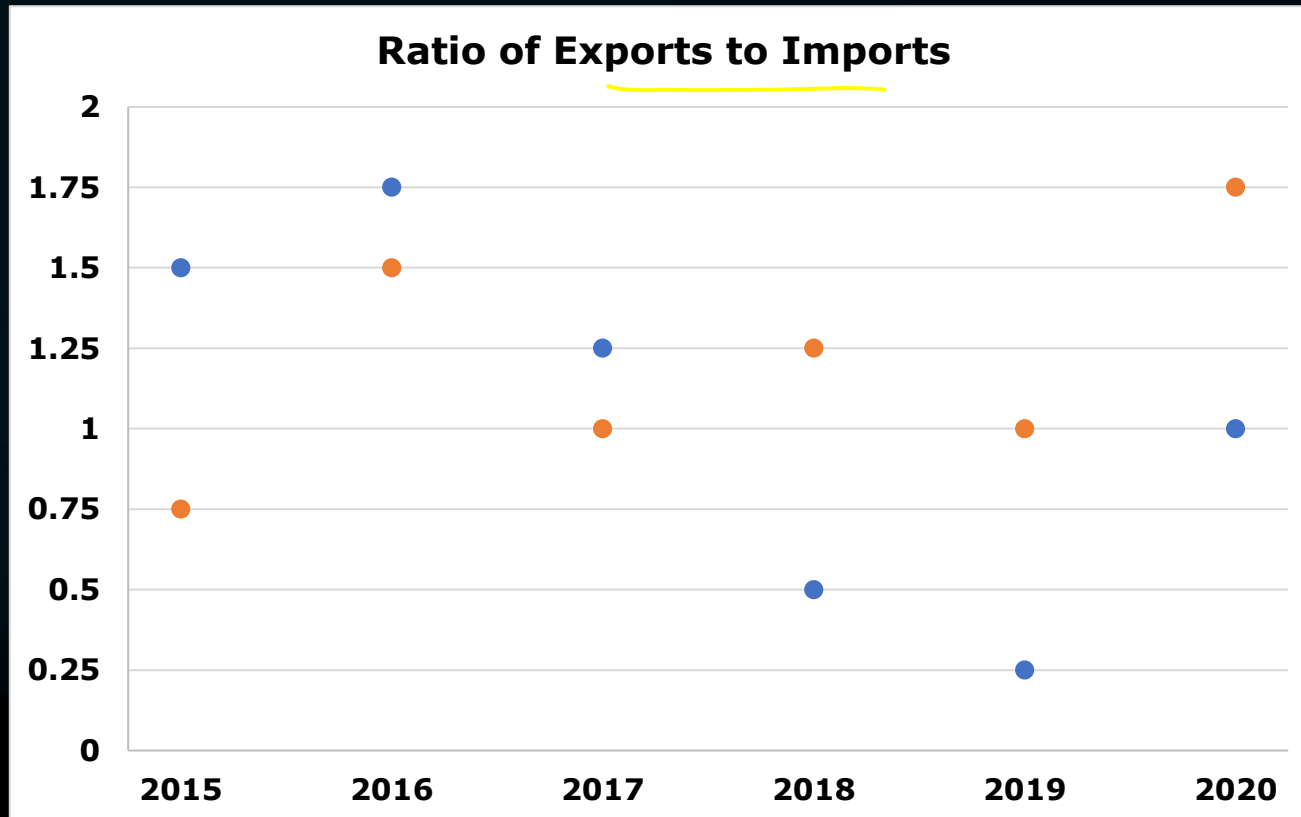
Set-1

Read the following data and answer the given questions.



The scatter chart given below shows the ratio of exports to imports(in terms of worth) of two companies P and Q over six years.(2015-2020)

Company P is shown in blue and company Q is shown in orange.



A. 1 **B. 3** **C. 4** **D. 2**

A. 2015 B. 2016 C. 2020 D. Cannot Be Determined

A. 140 crores B. 150 crores C. 180 crores D. 120 crores

A. 6 : 5 **B. 7 : 5** **C. 4 : 3** **D. 3 : 2**

A. 2 : 3 **B. 1 : 3** **C. 3 : 2** **D. 3 : 1**

QUESTION-1



#Q. In how many of the given years were the imports more than the exports for company Q?

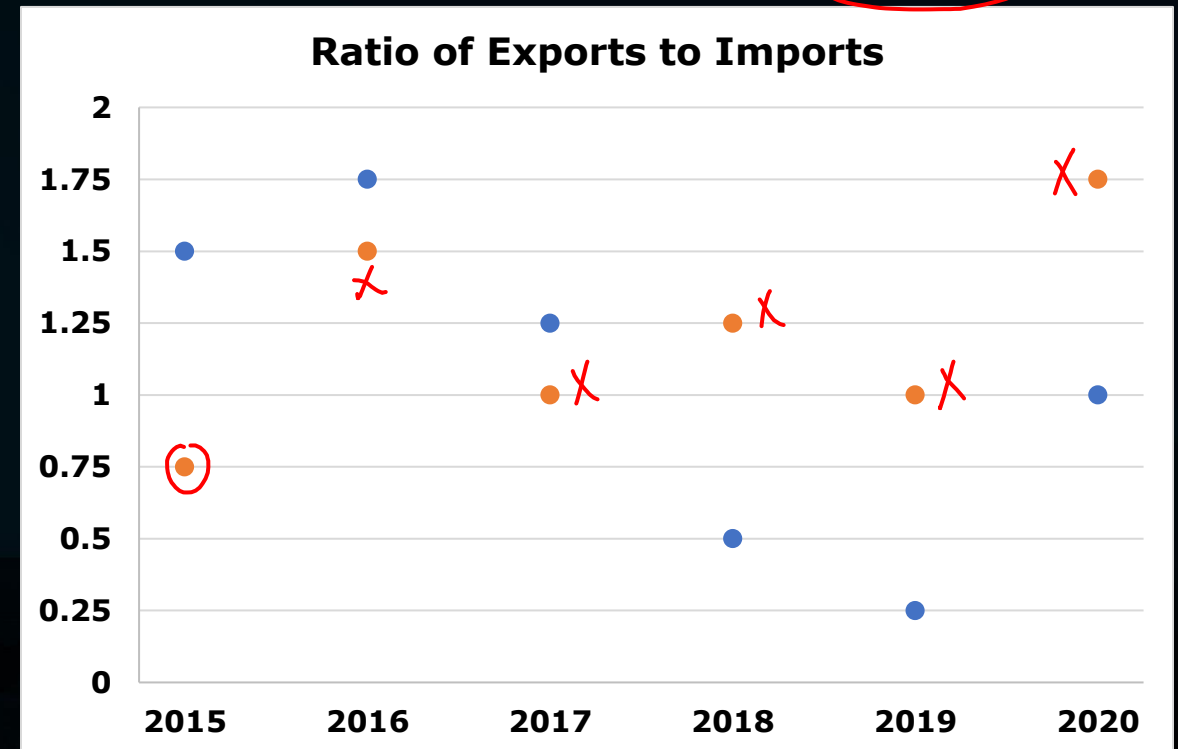
✓ A. 1
C. 4

B. 3
D. 2

$$I > E$$
$$I > \frac{E}{I}$$

$\frac{E}{I} < 1$

Company P : blue and company Q : orange.



QUESTION- 2



#Q. In which year was the difference between the amount of imports and exports of company Q the maximum:

A. 2015

B. 2016

C. 2020

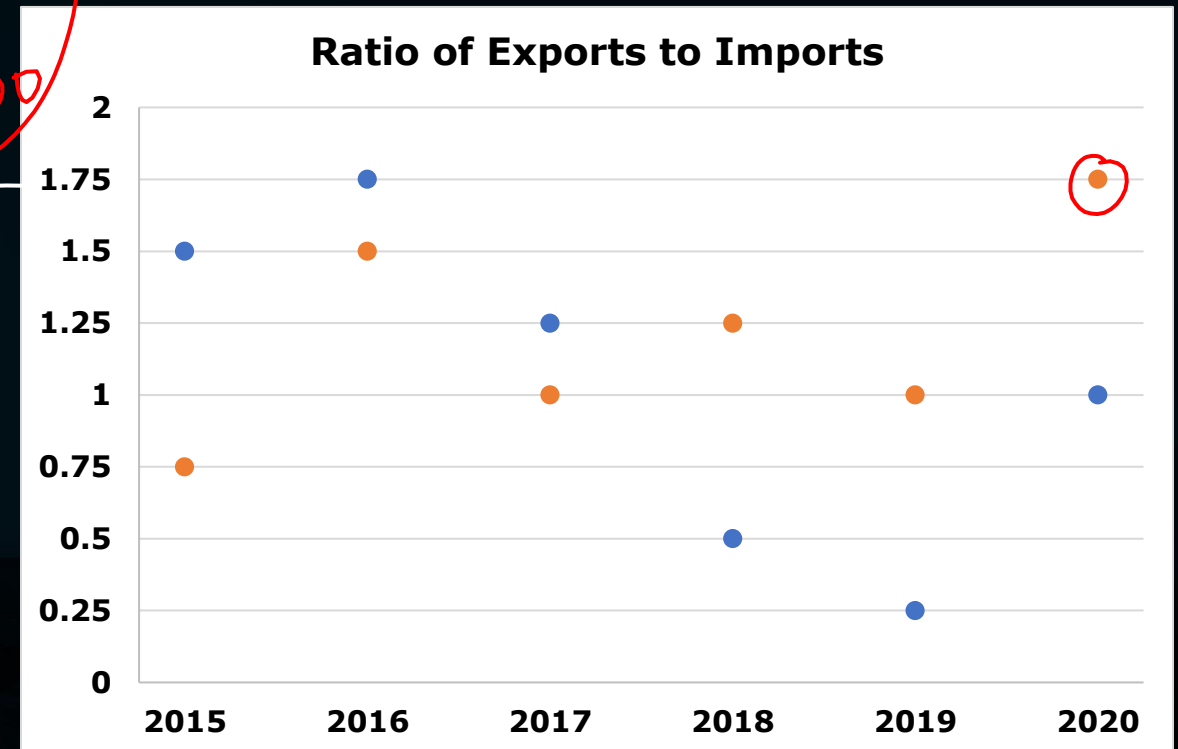
✓ D. Cannot Be Determined

$$\frac{2020}{E} = 1.75 = \frac{I}{4} \rightarrow \begin{matrix} 70 & 700 & 70,000 \\ 40 & 400 & 40,000 \end{matrix}$$

$$\frac{5}{4}$$

$$\frac{7}{4}$$

Company P : blue and company Q : orange.



QUESTION- 3



#Q. If the exports of company P in 2016 were Rs. 210 crores, what was the amount (in Rs.) of imports in that year?

~~A. 140 crores~~

B. 150 crores

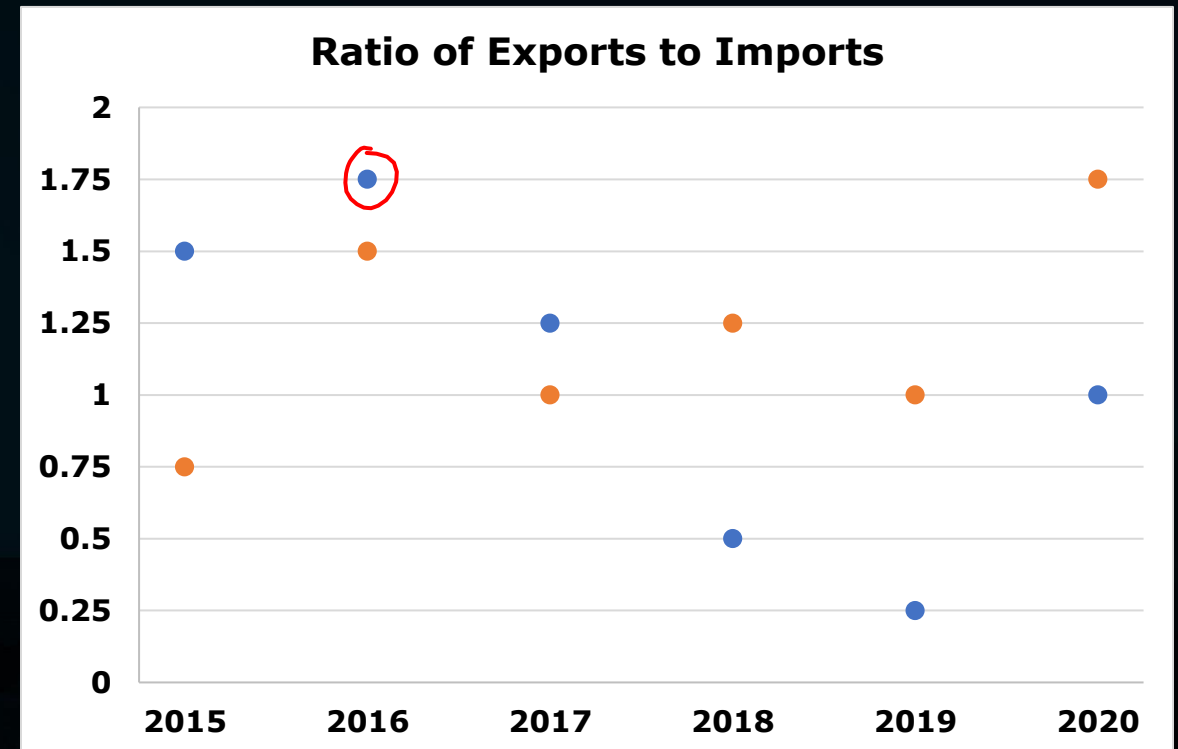
C. 180 crores

☒ D. 120 crores

$$\frac{E}{I} = 1.75 = \frac{7}{4} = \frac{210}{I}$$

$$I = 120$$

Company P : blue and company Q : orange.



QUESTION- 4



#Q. If the imports of company Q in 2020 were increased by 25%, then what would be the ratio of exports to the increased imports?

A. 6 : 5

✓ B. 7 : 5

C. 4 : 3

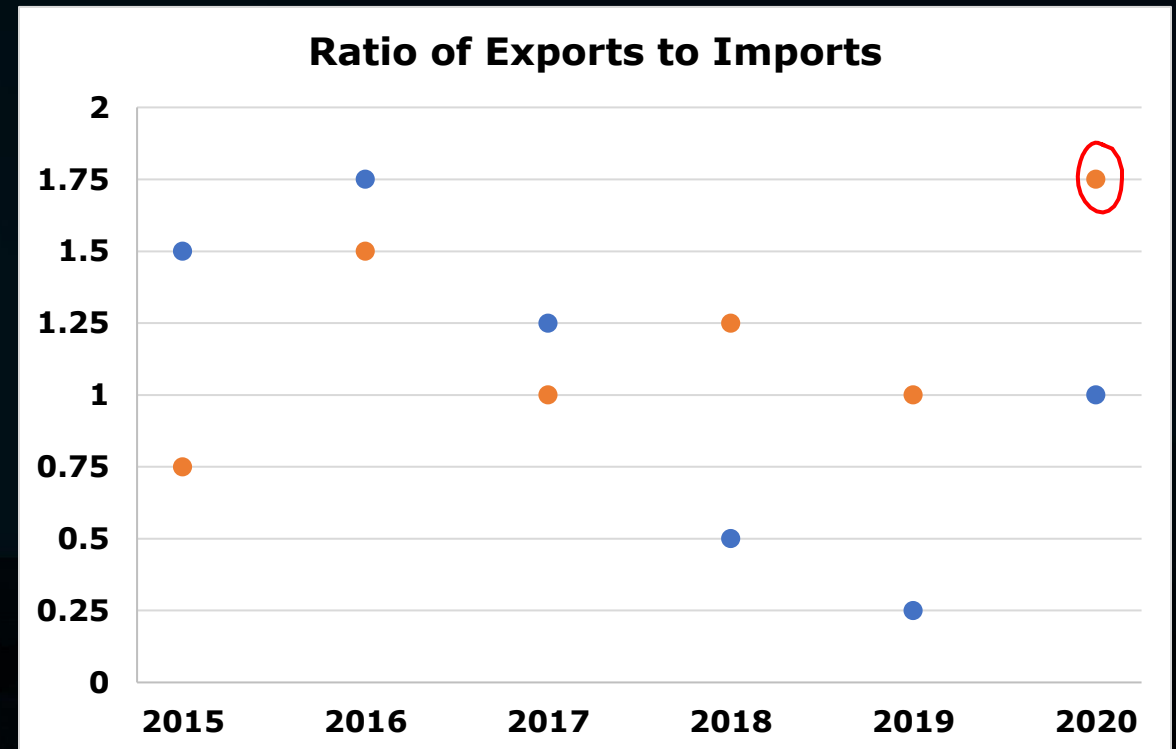
D. 3 : 2

$$\frac{E}{I} = 1.75 = \frac{7}{4}$$

$$\begin{aligned} E &= 7n \\ I &= 4n \end{aligned} \rightarrow 5n$$

$$\frac{7n}{5n} = \frac{7}{5}$$

Company P : blue and company Q : orange.



QUESTION- 5



#Q. What is the ratio of exports of Q in 2016 to the exports of P in 2018, if it is given that imports of Q in 2016 is equal to the imports of P in 2018?

A. 2 : 3

B. 1 : 3

C. 3 : 2

✓ D. 3 : 1

$$\frac{E_Q}{I_Q} = \frac{3}{2}$$

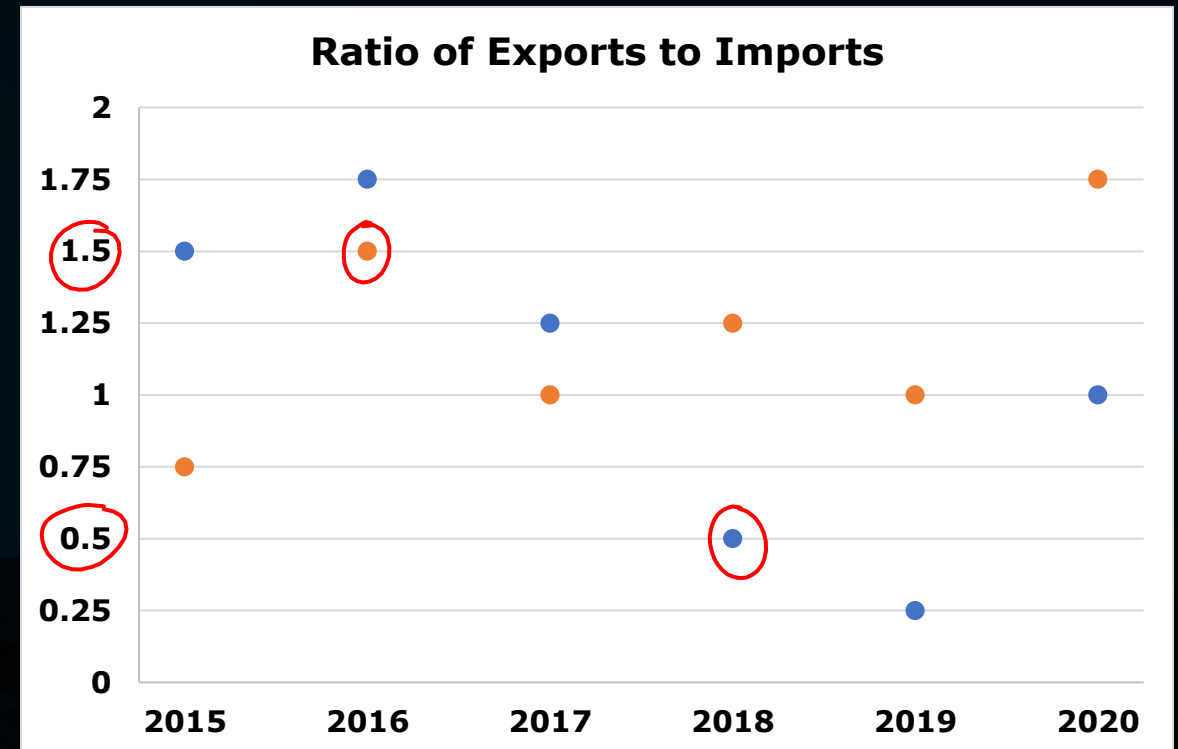
$$\frac{E_P}{I_P} = \frac{1}{2}$$

$$\frac{E_Q/I_Q}{E_P/I_P} = \frac{3/2}{1/2}$$

$$\frac{E_Q}{E_P} \times \frac{I_P}{I_Q} = \frac{3}{1}$$

$$\frac{E_Q}{E_P} = 3$$

Company P : blue and company Q : orange.





Set-2

Read the following data and answer the given questions.

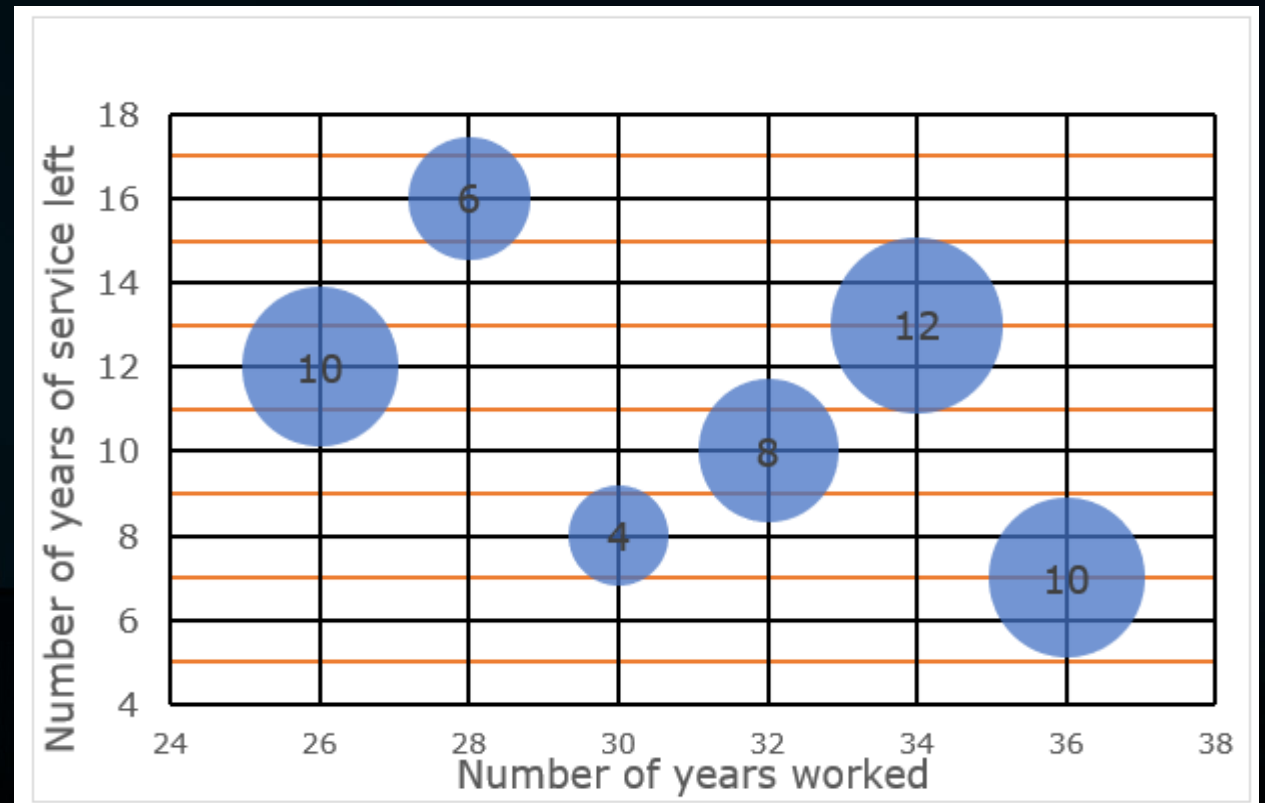


The bubble-chart given below shows the data of some of the senior employees of a company in year 2019. The employees were grouped according to the number of years worked by them in the company and the remaining number of years of their service, if they decided to work in the company till their retirement. The center of each bubble represents the number of years for which the employees have worked in the company (along the horizontal axis) and number of years of service left if they decide to work till retirement in the company (along the vertical axis).

The value inside the bubble gives the number of employees in the group.

It is known that the retirement age of each of the employee in the company was 65 years and none of the above grouped employees left their job after joining.

It was also known that all employees were born on the same day of a year and joined on the same day of the year. (not necessarily the same year)



QUESTIONS



#Q. How many employees were under the age of 23 at the time of joining?

- A. 22
- B. 26
- C. 28
- D. 34

#Q. What was the average joining age of all the given employees? (nearest integer)

- A. 21
- B. 22
- C. 23
- D. 24

#Q. If 40% of the given employees took early retirement and left at the age of 62 while the rest stayed till the usual retirement age, then what is the sum of the total number of years worked by all the given employees after year 2019?

- A. 464
- B. 494
- C. 504
- D. 654

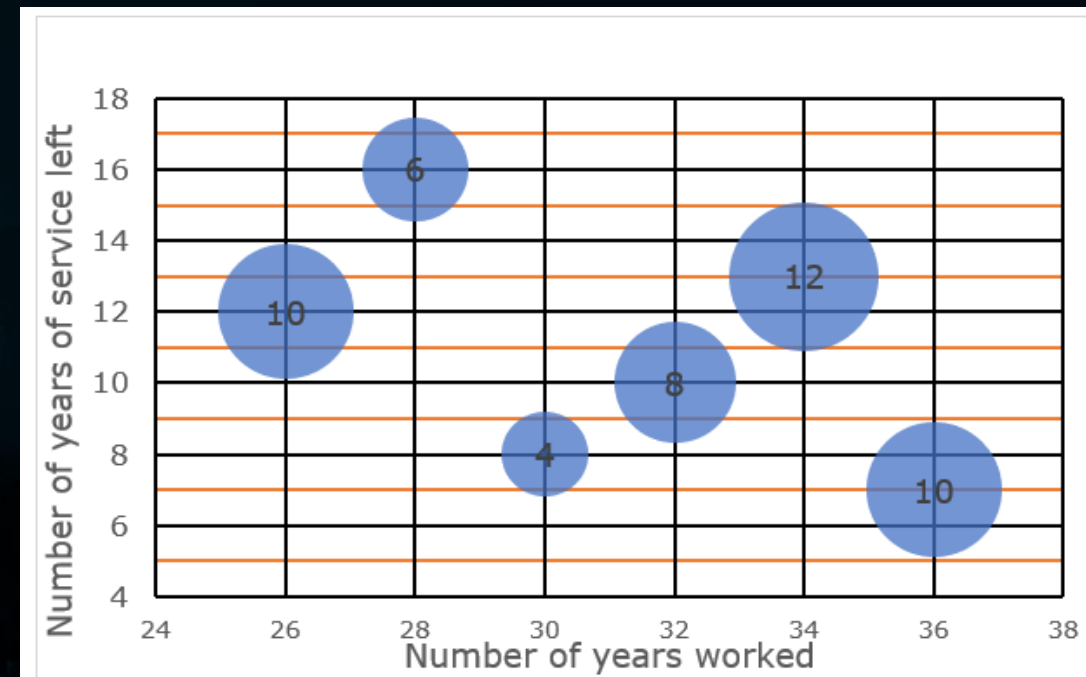
#Q. Considering 2019, how many employees have worked for more years than their age at the time of joining?

- A. 26
- B. 36
- C. 38
- D. 40

Analysis



	worked	left
10	26	12
6	28	16
4	30	8
8	32	10
12	34	13
10	36	7



QUESTION-1



#Q. How many employees were under the age of 23 at the time of joining?

A. 22

B. 26

☒ C. 28

D. 34

2019 (65)

	worked	left	Total	$65 - \uparrow = \text{Age}$
10	26	12	38	27
6	28	16	44	21
4	30	8	38	27
8	32	10	42	23
12	34	13	47	18
10	36	7	43	22

QUESTION- 2



- #Q. What was the average joining age of all the given employees? (nearest integer)
- A. 21
 B. 22
 C. 23
 D. 24

$$\text{Avg} = \frac{1124}{50} = 22.48$$

	worked	left	Total	65 - T = Age
10	26	12	38	27
6	28	16	44	21
4	30	8	38	27
8	32	10	42	23
12	34	13	47	18
10	36	7	43	22

$$\text{Sum} = 1124$$

$$N = 50$$

QUESTION- 3



#Q. If 40% of the given employees took ~~early retirement~~ and left at the age of 62 while the rest stayed till the usual retirement age, then what is the sum of the total number of years worked by all the given employees after year 2019?

A. 464

✓ B. 494

C. 504

D. 654

$$20 \times 3 = 60$$

	worked	left	Total	65 - ↑ = Age
50 ↑ 10	26	12	38	27
6	28	16	44	21
4	30	8	38	27
8	32	10	42	23
12	34	13	47	18
10	36	7	43	22

$$\text{Sum} = 554 - 60 = 494$$

QUESTION- 4



#Q. Considering 2019, how many employees have worked for more years than their age at the time of joining?

A. 26

B. 36

C. 38

✓ D. 40

	worked	left	Total	$65 - \uparrow = \text{Age}$	
✓ 10	26	12	38	27	X
6	28	16	44	21	✓
4	30	8	38	27	✓
8	32	10	42	23	✓
12	34	13	47	18	✓
10	36	7	43	22	✓

$$50 - 10 = 40$$

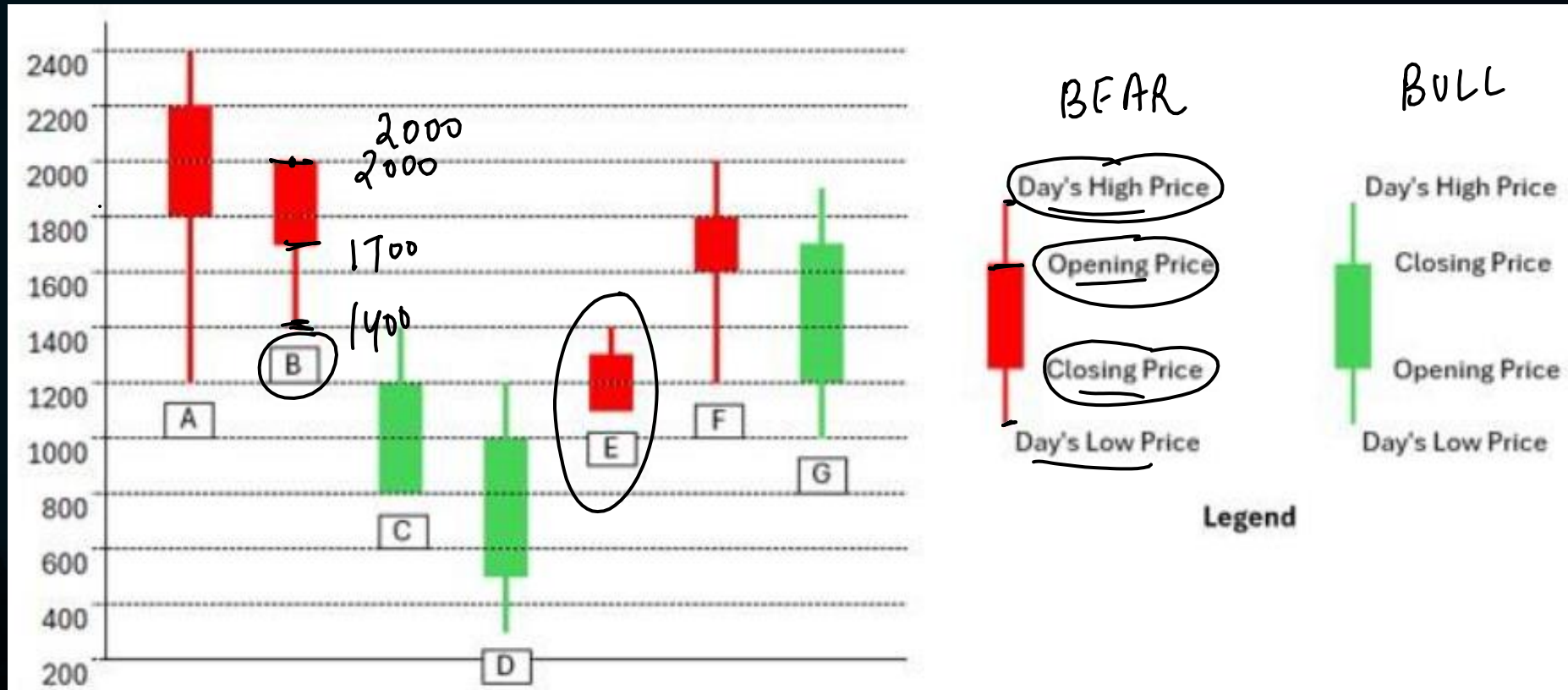


Set-3

Read the following data and answer the given questions.



The chart below shows the price data for seven shares – A, B, C, D, E, F, and G as a candlestick plot for a particular day. The vertical axis shows the price of the share in rupees. A share whose closing price (price at the end of the day) is more than its opening price (price at the start of the day) is called a bullish share; otherwise, it is called a bearish share. All bullish and bearish shares are shown in green and red colour respectively.



QUESTION-1



#Q. Daily Share Price Variability (SPV) is defined as (Day's high price - Day's low price) / (Average of the opening and closing prices during the day). Which among the shares A, C, D and F had the highest SPV on that day?

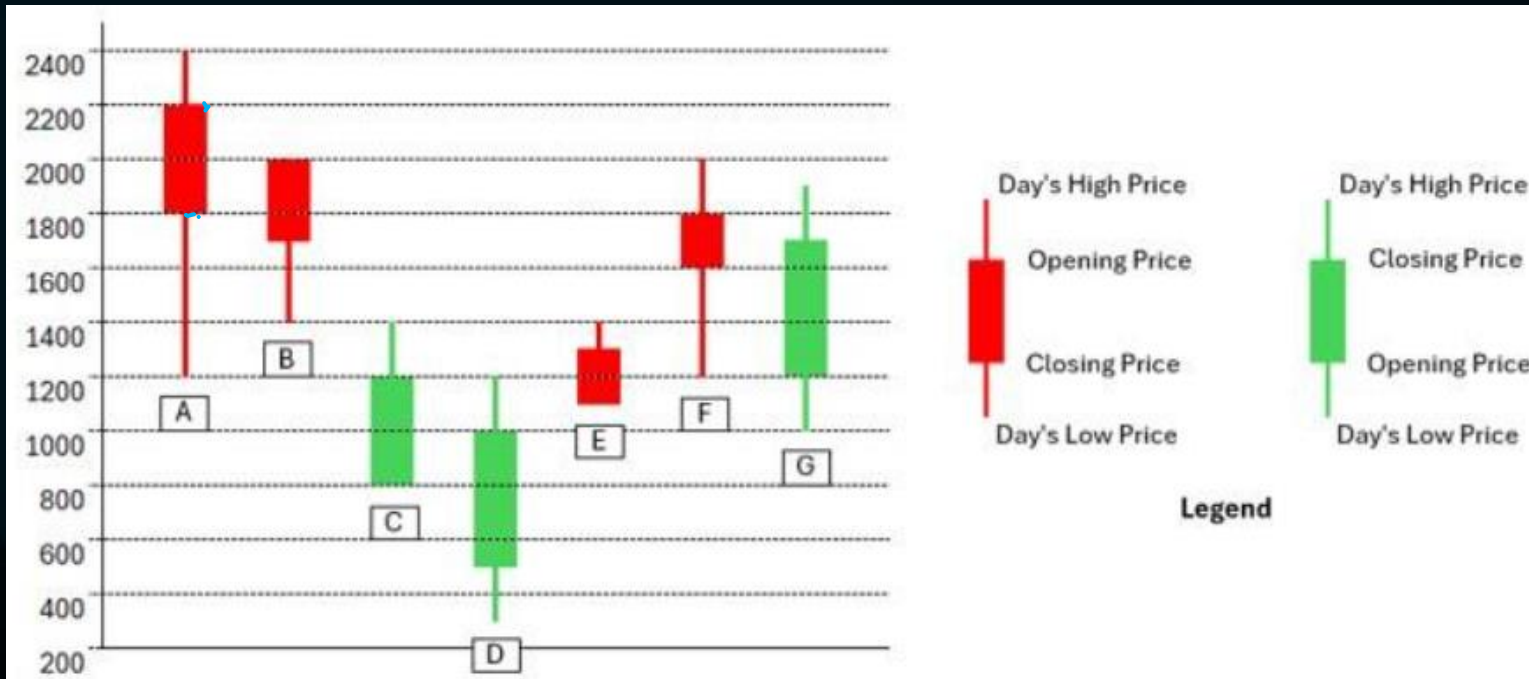
- a. A b. F ☒ c. D d. C

$$A = \frac{1200}{2000} \checkmark$$

$$C = \frac{600}{1000} \checkmark$$

$$D = \frac{900}{750} \checkmark$$

$$F = \frac{800}{1700} \times$$



QUESTION- 2

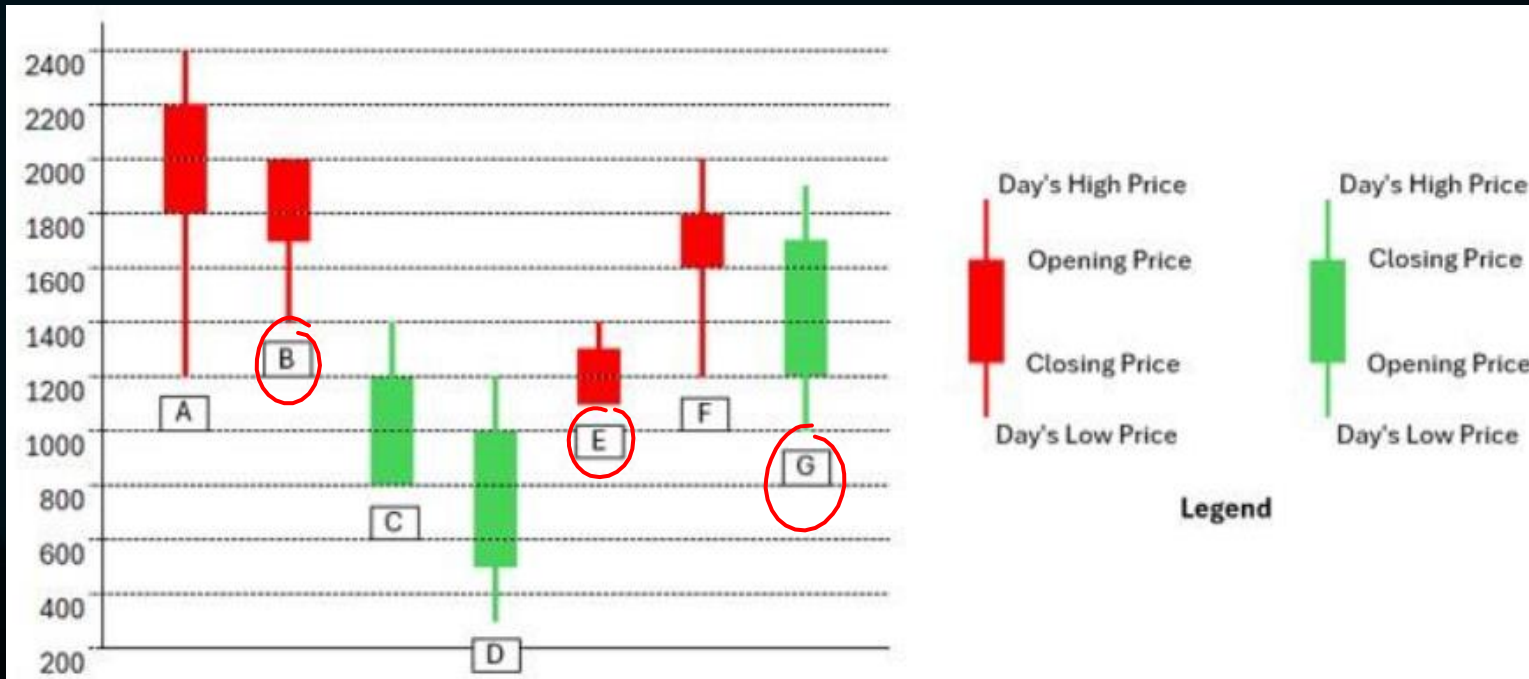


#Q. Daily Share Price Variability (SPV) is defined as (Day's high price - Day's low price) / (Average of the opening and closing prices during the day). How many shares had an SPV greater than 0.5 on that day?

$$B = \frac{600}{1850} \times$$

$$E = \frac{300}{1200} \times$$

$$G = \frac{900}{1450} \checkmark \checkmark$$



A C D G

4

QUESTION- 3



#Q. Daily loss for a share is defined as (Opening price – Closing price) / (Opening price). Which among the shares A, B, F and G had the highest daily loss on that day?

- a. B b. G c. F d. A

$$A = \frac{400}{2200} = \frac{1}{5.5}$$

$$B = \frac{300}{2000} = \frac{1}{6.66}$$

$$F = \frac{200}{1800} = \frac{1}{9}$$

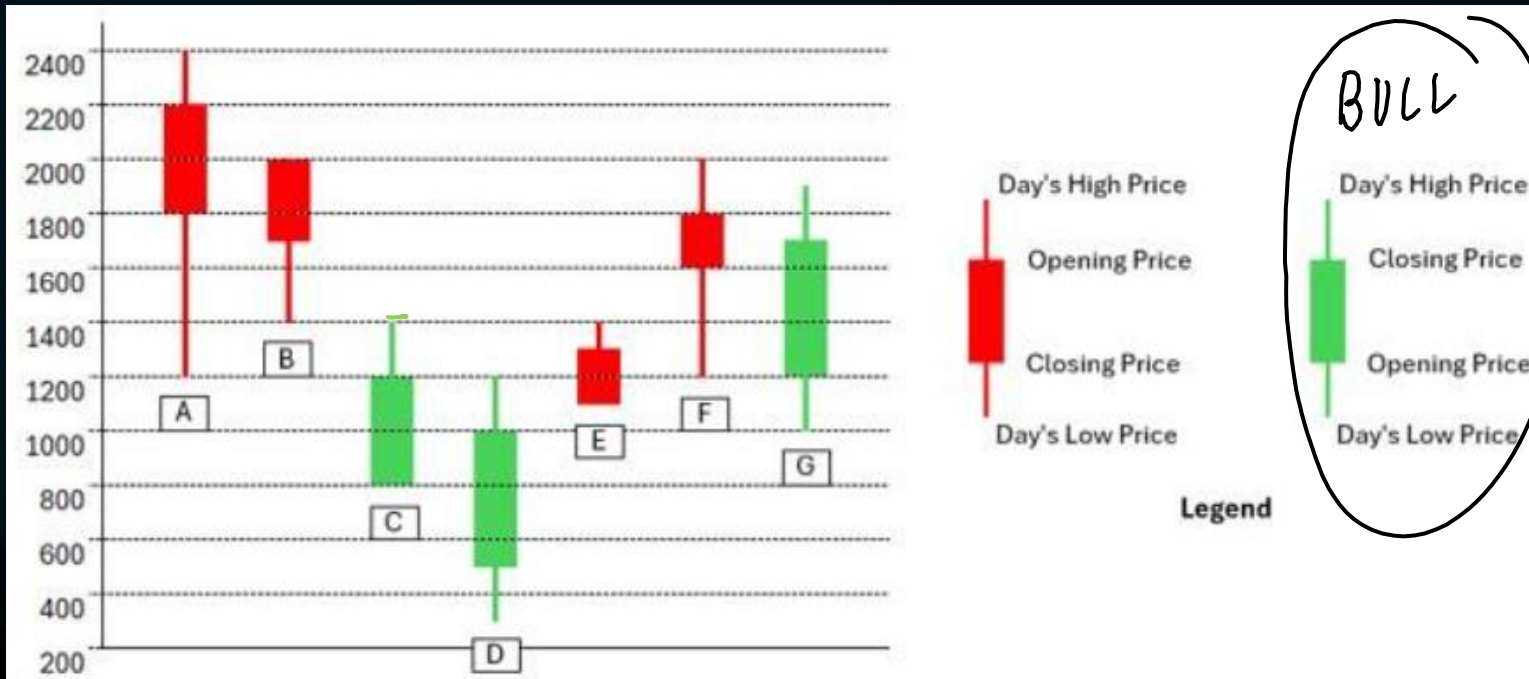
G = Negative X



QUESTION- 4



- #Q. What would have been the percentage wealth gain for a trader, who bought equal numbers of all bullish shares at opening price and sold them at their day's high?
- a. 100% ☒ b. 80% c. 72% d. 50%



	Buy	Sell
C	800	1400
D	500	1200
G	1200	1900
	2500	4500
	80%	



THANK
You

